INCH-POUND
MIL-PRF-3098/118E
11 July 2011
SUPERSEDING
MIL-PRF-3098/118D
27 July 2005

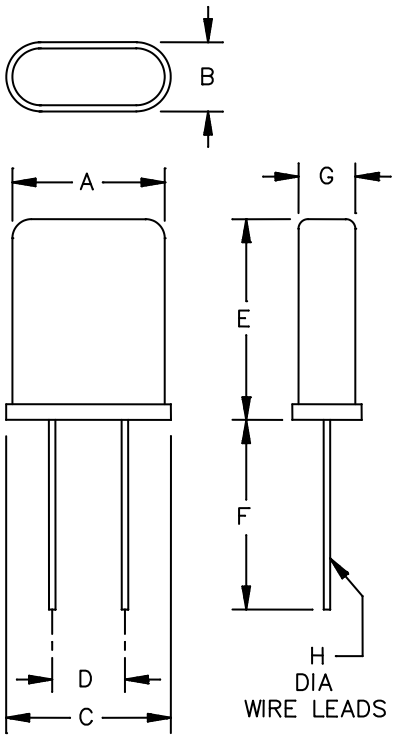
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR139/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 20 MHz to 22 MHz; fundamental; noncontrolled; antiresonance.



Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.402	---	10.21
B	---	.183	---	4.65
C	---	.435	---	11.05
D	.184	.200	4.67	5.08
E	---	.530	---	13.46
F	.50	---	12.7	---
G	---	.150	---	3.81
H	.015	.019	0.38	0.48

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR139/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 20 MHz to 22 MHz, inclusive.

Frequency tolerance: ±50 ppm.

Mode of oscillation: Fundamental.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Capacitance, shunt: 12.0 pF, maximum.

Antiresonance load capacitance: 30.0 pF ±0.5 pF.

Equivalent resistance: 40 ohms, maximum.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Part or Identifying Number (PIN): For Product Level B, PIN shall be CR139/U or CR139B/U (followed by specified frequency). For Product Level S, PIN shall be CR139S/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-PRF-202

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project 5955-2011-007)

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